P1 $\varepsilon_r(\omega) = 1 - \frac{\omega_p^2}{\omega^2}$

(a) $\frac{\omega}{c/n} = K$ $\quad n = \sqrt{\frac{\varepsilon(\omega)\mu(\omega)}{\varepsilon_0 \mu_0}} = \sqrt{\varepsilon_r(\omega)} = \sqrt{1 - \frac{\omega_p^2}{\omega^2}}$

$\Rightarrow \quad \omega^2 n^2 = c^2 K^2, \quad \omega^2\left(1 - \frac{\omega_p^2}{\omega^2}\right) = c^2 K^2$

$$\omega^2 = \omega_p^2 + c^2 K^2$$

$$\omega = \sqrt{\omega_p^2 + c^2 K^2}.$$

Phase velocity $v_{ph} = \frac{c}{n} = c\sqrt{\frac{\omega}{\omega^2 - \omega_p^2}} > c$

Group velocity $v_g = \frac{d\omega}{dK} = \frac{c^2 K}{\sqrt{\omega_p^2 + c^2 K^2}} = c\,\frac{\sqrt{\omega^2 - \omega_p^2}}{\omega} =$

$$= c\sqrt{1 - \frac{\omega_p^2}{\omega^2}} < c$$

$v_{ph} \approx v_g$ when $\frac{\omega_p}{\omega} \to 0$, i.e. for $\omega \gg \omega_p$.

(b) $t_1 = \frac{L}{v_g(\omega_1)}, \quad t_2 = \frac{L}{v_g(\omega_2)}$

$\Rightarrow \quad t_1 v_g(\omega_1) = t_2 v_g(\omega_2)$

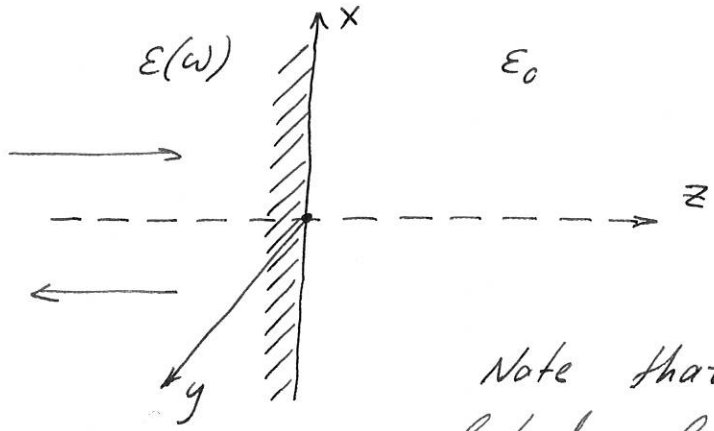
$$t_1^2\left(1 - \frac{\omega_p^2}{\omega_1^2}\right) = t_2^2\left(1 - \frac{\omega_p^2}{\omega_2^2}\right)$$

$$\omega_p^2\left(\frac{t_2^2}{\omega_2^2} - \frac{t_1^2}{\omega_1^2}\right) = t_2^2 - t_1^2$$

$$\omega_p^2 = \frac{t_2^2 - t_1^2}{\frac{t_2^2}{\omega_2^2} - \frac{t_1^2}{\omega_1^2}} = \omega_1^2\omega_2^2\,\frac{t_2^2 - t_1^2}{t_2^2\omega_1^2 - t_1^2\omega_2^2}$$

$$\omega_p^2 = \frac{N_e e^2}{\varepsilon_0 m_e} \Rightarrow N_e = \frac{\varepsilon_0 m_e}{e^2}\,\frac{\omega_1^2\omega_2^2(t_2^2 - t_1^2)}{t_2^2\omega_1^2 - t_1^2\omega_2^2}.$$

Problem 2



Consider an incident wave polarized along x-direction:

$$\hat{x} E_0 e^{-i\omega t + ikz}$$

Note that ω and k are related by

$$\frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0} = k^2$$

as $\frac{\varepsilon(\omega)}{\varepsilon_0} \equiv \epsilon_r = 1 - \frac{\omega_p^2}{\omega^2} \Rightarrow \boxed{\omega^2 - \omega_p^2 = c^2 k^2}$ (1)

Electric field (only x-components are non-zero) is

$$E_x(z<0) = E_0 e^{-i\omega t + ikz} + E_R e^{-i\omega t - ikz} \quad (2)$$

$$E_x(z>0) = E_t e^{-i\omega t + i\tilde{k} z} \quad (3)$$

where $\tilde{k}$ is related to ω via simply $\omega = c\tilde{k}$ (4)

Magnetic field is found from

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$$ and is

along the y-direction:

$$B_y = -\frac{i}{\omega}(\nabla \times \vec{E})_y = -\frac{i}{\omega}\frac{\partial E_x}{\partial z}$$

with the help of Eqs. (2) – (3)

$$B_y(z<0) = \frac{k}{\omega} E_0 e^{-i\omega t + ikz} - \frac{k}{\omega} E_R e^{-i\omega t - ikz} \quad (5)$$

$$B_y(z>0) = \frac{\tilde{k}}{\omega} E_t e^{-i\omega t + i\tilde{k}z} \quad (6)$$

Continuity of E_x and B_y at $z=0$ gives

$$E_0 + E_R = E_t$$

$$k E_0 - k E_R = \tilde{k} E_t$$

which yield E_R:

$$E_R = \frac{k - \tilde{k}}{k + \tilde{k}} E_0$$

and, since both k and $\tilde{k}$ are real for $\omega > \omega_p$

$$R = \left|\frac{E_R}{E_0}\right|^2 = \left(\frac{k - \tilde{k}}{k + \tilde{k}}\right)^2$$

Using expressions for k and $\tilde{k}$ we finally obtain:

$$R = \left(\frac{\omega - \sqrt{\omega^2 - \omega_p^2}}{\omega + \sqrt{\omega^2 - \omega_p^2}}\right)^2$$

when $\omega \gg \omega_p$ the reflection coefficient approaches zero: $\sqrt{\omega^2 - \omega_p^2} \approx \omega - \frac{\omega_p^2}{2\omega}$

and $$R \approx \left(\frac{\omega_p^2/2\omega}{2\omega}\right)^2 \approx \frac{\omega_p^2}{16\omega^2}$$

When $\omega \to \omega_p$ the reflection coefficient approaches 1:

for $\omega = \omega_p - \varepsilon$ where $\varepsilon \ll \omega_p$:

$$\sqrt{\omega^2 - \omega_p^2} \approx \sqrt{2\omega_p \varepsilon}$$

and

$$R \approx \left(\frac{\omega_p - \sqrt{2\omega_p \varepsilon}}{\omega_p + \sqrt{2\omega_p \varepsilon}} \right)^2 \approx 1 - 4\sqrt{\frac{\varepsilon}{\omega_p}}$$